

SNEHA MURALI

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U.S. Citizen

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Education

University of Washington

Sep 2023 – June 2027

Bachelor of Science in Computer Science, Minor in Data Science

Seattle, WA

- **Relevant Coursework:** Data Structures, Software Design, Algorithms, Databases, Hardware/Software Interface, Mobile App Development, Computer Security, human computer interaction

Technical Skills

Languages: Java, Python, SQL, R, JavaScript, Dart, Flutter, C, C++,

Developer Tools: React, HTML, Scikit-learn, Pandas, NumPy, Git, Microsoft Excel, Figma, Tableau

Technologies/Frameworks: Data Structures, object oriented programming, full-stack development

Experience

FridgeFinder | Backend Developer, Python, AWS DynamoDB, AWS Lambda

July 2025 – Ongoing

- Designed and implemented a DynamoDB-backed user rewards system in Python using Boto3, enabling atomic updates to user points and engagement metrics with concurrency-safe writes and strong consistency guarantees.
- Built and deployed scalable backend APIs using AWS Lambda and DynamoDB to aggregate, classify, and serve real-time community fridge status data for a production application (*fridgefinder.app*).
- Engineered robust error-handling and data validation logic to process incomplete, inconsistent, and unexpected real-world user reports, improving backend reliability and observability.
- Developed unit and integration tests using Moto to mock AWS services and validate DynamoDB operations, ensuring correctness and reliability of backend functionality.
- Collaborated in an Agile development environment using GitHub, contributing pull requests, code reviews, and iterative improvements to maintain production code quality.

UW Buckley Lab

March 2026 - Ongoing

Undergraduate Researcher

Seattle, WA

- Built Python pipelines to retrieve butterfly specimen images and associated metadata from museum and biodiversity APIs (iDigBio, GBIF, Smithsonian Open Access, Natural History Museum CKAN API), implementing API querying, nested JSON parsing, metadata standardization, and automated image downloading across large museum collections.
- Executing and validating Python-based image analysis pipelines to quantify butterfly coloration across large datasets.
- Running, debugging, and monitoring terminal-based workflows; contributing to the development of automated data pipelines using Python and GitHub.

UW School of Medicine (SOM)

Feb 2026 – Ongoing

Web Developer Volunteer

Seattle, WA

- Developed and maintained Firebase/Firestore-backed features for the Service Learning website, including volunteer event sign-up workflows, attendee tracking, and admin dashboards for project management.
- Designed and implemented role-based authorization logic for admins and project leads using Firebase Auth and Firestore collections to securely manage volunteer participant information. Developed backend features to query, aggregate, and display participant attendance data for admin-facing views, improving visibility into volunteer engagement across events.
- Selected to serve as **Lead Developer** for the 2026–2027 academic year, overseeing development priorities and supporting a student-led engineering team.

Comtech Telecommunications Corp. — Software Engineering Intern (Incoming)

June 2026

Seattle, WA

Projects

Migraine Tracking App | *Flutter, APIs*

May – June 2025

- * Developed a **cross-platform mobile app** using **Flutter** to track migraine symptoms and triggers.
- * Integrated the **iOS Health API** and built **data visualizations** for symptom tracking.
- * Built reusable UI components and implemented **state management**.

File System Search Engine | *C, C++, Data Structures, Systems Programming*

June - August 2025

- * Built a **file system crawler, inverted index, and console-based search engine** in **C/C++** to recursively index and query text documents.
- * Implemented core data structures including **hash tables, linked lists, iterators, and posting lists** to support efficient document indexing and ranked search.
- * Engineered a **query processor** supporting multi-word search and relevance ranking based on document term frequency.
- * Debugged memory and pointer issues using **Valgrind** and developed modular, testable systems code under strict performance and correctness requirements.

Traffic Congestion Optimization | *Python*

Dec 2023

- * Developed a **neural network model** reducing congestion predictions by **15%**.
- * Integrated external APIs and iterated under time constraints during a fast-paced hackathon environment.
- * Awarded **Honorable Mention** at the **INRIX × Amazon Hackathon**.

Gestational Diabetes Prediction Model | *Python, Scikit-learn*

July – Aug 2024

- * Developed a **logistic regression model** achieving **85% accuracy** for gestational diabetes prediction.
- * Performed **feature engineering** and identified heredity as the strongest predictor.
- * Evaluated model performance using **ROC curves, AUC**, and confusion matrices.

Music Match | *Python, AWS, PostgreSQL*

Jan – Mar 2024

- * Built a **music recommendation system** using the **Spotify API, AWS EC2**, and **PostgreSQL**.
- * Collaborated in **CSEED Cohort 0** to design and present a functional prototype.